

ENTHUSIAST ADVANCE COURSE

PHASE : MEF-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 14-08-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	4	2	3	3	1	2	4	2	4	1	4	3	2	2	2	2	1	2	3	3	3	3	1	2	4	1	2	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	2	2	3	3	1	4	3	2	1	4	3	3	3	4	1	4	4	3	1	3	3	3	2	4	3	4	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	4	1	1	2	3	2	2	3	2	3	3	1	4	3	1	4	1	4	1	1	4	1	3	1	2	1	2	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	2	2	2	2	3	4	2	2	3	4	1	3	3	1	2	1	2	3	1	2	1	4	2	2	4	4	2	4	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	2	3	1	4	4	2	4	4	2	3	4	2	4	4	2	4	3	4	4	3	1	2	4	3	3	1	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	2	4	3	4	4	3	2	2	1	2	4	3	3	3	4	4	3	3	4	4	4	3	3	1	3	1	3	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	4	2	3	3	2	2	4	3	4	4	4	4	4	2	2	4	4	3	3	3										

HINT - SHEET

SUBJECT : PHYSICS

SECTION - A

1. Ans (1)

For retracing of light, light will fall normally on the mirror.

By applying Snell's law $\mu_1 \sin i = \mu_2 \sin r$

$$1 \times \sin 45^\circ = \mu \sin 30^\circ$$

$$\mu = \sqrt{2}$$

2. Ans (4)

$$m = \frac{f}{f - u}$$

Image to be real and three times $m = -3$

$$-3 = \frac{f}{f - (-20)}$$

$$f = -15 \text{ cm}$$

3. Ans (2)

Object is in rare medium & observed from denser

$$d_{app} = \mu d_{ac}$$

$$d_{ac} = \frac{d_{app}}{\mu} = \frac{30}{4/3} \Rightarrow \frac{90}{4} \Rightarrow 22.5 \text{ cm}$$

4. Ans (3)

$$P_{eq} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$= \frac{100}{(+40)} + \frac{100}{-25}$$

$$= (2.5) - 4 = -1.5 \text{ D}$$

5. Ans (3)

An observer can see a rainbow when his back is towards the sun.

6. Ans (1)

Focal length of a mirror doesn't depend on surrounding medium.

7. **Ans (2)**

$$P_1 = P \quad P_2 = \frac{P}{2} \quad P_3 = \frac{P}{2}$$

8. **Ans (4)**

$$\text{Microscope} \rightarrow M \propto \frac{1}{f_0}$$

$$\text{Telescope} \rightarrow M \propto f_0$$

9. **Ans (2)**

$$n = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin \frac{A}{2}}$$

$$\Rightarrow \sqrt{3} = \frac{\sin\left(\frac{A+A}{2}\right)}{\sin \frac{A}{2}}$$

$$\Rightarrow \sqrt{3} \sin \frac{A}{2} = \sin A$$

$$\Rightarrow \sqrt{3} \sin \frac{A}{2} = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \cos \frac{A}{2}$$

$$\Rightarrow \frac{A}{2} = 30^\circ$$

$$\Rightarrow A = 60^\circ$$

10. **Ans (4)**

$$u \rightarrow \infty$$

$$V = -50 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{-1}{50} - \frac{1}{\infty}$$

$$\Rightarrow f = -50 \text{ cm} = -0.5 \text{ m}$$

$$\therefore P = \frac{1}{-0.5} D$$

$$\Rightarrow P = -2D$$

11. **Ans (1)**

$$v^2 = u^2 + 2gh$$

12. **Ans (4)**

$$v = \frac{ds}{dt} = \beta + 2\gamma t + 4\delta t^3 = \beta \text{ (at } t = 0)$$

$$a = \frac{dv}{dt} = 2\gamma + 12\delta t^2 = 2\gamma \text{ (at } t = 0)$$

$$\text{Ratio} = \frac{\beta}{2\gamma}$$

13. **Ans (3)**

TG: @Chalnaayaaar

In uniform circular motion speed is constant but velocity varies.

14. **Ans (2)**

Total distance travelled =

$$60 \times 1 + 40 \times \frac{1}{2} = 80 \text{ km}$$

$$\text{and Total time taken} = 1 \text{ hr} + \frac{1}{2} \text{ hr} = \frac{3}{2} \text{ hr}$$

$$\therefore \text{Average speed} = \frac{80}{3/2} = 53.33 \text{ km/h}$$

15. **Ans (2)**

Magnetic field inside the cylindrical conductor

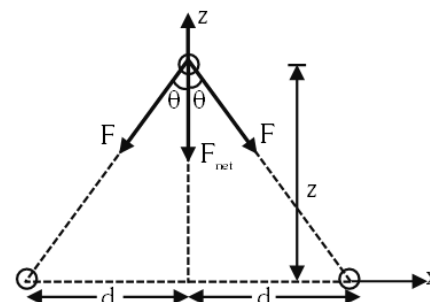
$$B_{\text{in}} = \frac{\mu_0}{4\pi} \cdot \frac{2ir}{R^2} \quad (R = \text{Radius of cylinder, } r = \text{distance of observation point from axis of cylinder})$$

Magnetic field out side the cylinder at a distance r'

$$\text{from it's axis } B_{\text{out}} = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r'}$$

$$\Rightarrow \frac{B_{\text{in}}}{B_{\text{out}}} = \frac{rr'}{R^2} \Rightarrow \frac{10}{B_{\text{out}}} = \frac{\left(R - \frac{R}{4}\right)(R + 4R)}{R^2}$$

16. **Ans (2)**

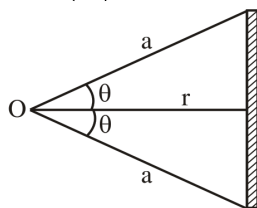


$$F = \frac{\mu_0 i^2}{2\pi \sqrt{d^2 + z^2}}$$

$$F_{\text{Net}} = 2F \cos \theta = 2 \frac{\mu_0 i^2}{2\pi \sqrt{d^2 + z^2}}$$

$$= \frac{z}{\sqrt{d^2 + z^2}} = \frac{\mu_0 i^2 z}{\pi(d^2 + z^2)}$$

17. Ans (2)



$$r = a \cos \theta$$

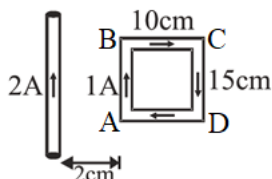
$$\begin{aligned} B_0 &= \frac{\mu_0 i}{4\pi r} (\sin \theta + \sin \theta) \\ &= \frac{\mu_0 i}{2\pi} \times \frac{\sin \theta}{a \cos \theta} \\ &= \frac{\mu_0 i}{2\pi a} \tan \theta \end{aligned}$$

due to n sides,

$$\begin{aligned} B_{\text{net}} &= nB_0 \\ &= n \left(\frac{\mu_0 i}{2\pi a} \tan \frac{\pi}{n} \right) \end{aligned}$$

18. Ans (1)

Force on side BC and AD are equal but opposite so their net will be zero.



$$\text{But } F_{AB} = 10^{-7} \times \frac{2 \times 2 \times 1}{2 \times 10^{-2}} \times 15 \times 10^{-2} = 3 \times 10^{-6} \text{ N}$$

$$\begin{aligned} \text{and } F_{CD} &= 10^{-7} \times \frac{2 \times 2 \times 1}{(12 \times 10^{-2})} \times 15 \times 10^{-2} \\ &= 0.5 \times 10^{-6} \text{ N} \end{aligned}$$

$$\begin{aligned} \Rightarrow F_{\text{net}} &= F_{AB} - F_{CD} = 2.5 \times 10^{-6} \text{ N} \\ &= 25 \times 10^{-7} \text{ N, towards the wire.} \end{aligned}$$

19. Ans (2)

$$\vec{B} \cdot \vec{a} = 0$$

$$2x - 6 + 4 = 0$$

$$2x = 2$$

$$x = 1$$

20. Ans (3)

Here the proton has no acceleration so $E = B = 0$. When $E = 0$ but $B \neq 0$, but parallel to the motion of proton, there will be no force acting. When $E \neq 0$ and $B \neq 0$ and E, B and motion of proton (v) are mutually perpendicular, there may be no net force. forces due to E and B cancel each other.

21. Ans (3)

TG: @Chalnaayaaar

The magnetic field at the centre of a circle is given by

$$B = \frac{\mu_0 i}{2r}$$

where, i is current and r the radius of circle.

$$\text{Also, } i = \frac{q}{t}$$

For helium nucleus, $q = 2e$

$$\therefore i = \frac{2e}{t}$$

$$\begin{aligned} \text{so, } B &= \frac{\mu_0 \cdot 2e}{2rt} \\ &= \frac{\mu_0 \times 2 \times 1.6 \times 10^{-19}}{2 \times 0.8 \times 2} = 10^{-19} \mu_0 \end{aligned}$$

22. Ans (3)

$$\vec{a} = \frac{q}{m} [\vec{E} + (\vec{v} \times \vec{B})]$$

$$\text{When released from rest } \vec{a} = \frac{e}{m} [\vec{E}] = a_0 \hat{j}_{(\text{north})}$$

$$\Rightarrow \vec{E} = \frac{ma_0}{e} \hat{j} (\text{North})$$

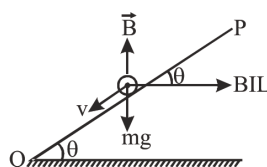
When project in east with speed v_0

$$\vec{a} = \frac{e}{m} \left[\frac{ma_0}{e} \hat{j} + (v_0 \hat{i} \times \vec{B}) \right] = 3a_0 (-\hat{j}) (\text{south})$$

$$\Rightarrow v_0 (\hat{i} \times \vec{B}) = -\frac{4ma_0}{e} \hat{j} \Rightarrow \vec{B} = \frac{4ma_0}{ev_0} (\hat{k}) (\text{up})$$

23. Ans (3)

Magnetic force on the rod $F_m = BIL$. It acts in the direction as shown in figure. The rod will move with a constant speed if the net force on the rod is zero. It will be so if



$$BIL \cos \theta = mg \sin \theta$$

$$\text{or } B = \frac{mg}{IL} \tan \theta$$

27. Ans (1)

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M_w}}$$

$$\frac{(V_{\text{rms}})_{\text{He}}}{(V_{\text{rms}})_{\text{H}_2}} = \sqrt{\frac{T_{\text{He}}}{T_{\text{H}_2}} \times \frac{M_{\text{WH}_2}}{M_{\text{WH}_e}}}$$

$$\frac{1}{\sqrt{2}} = \sqrt{\frac{T_{\text{He}}}{273} \times \frac{2}{4}}$$

$$\frac{1}{2} = \frac{T}{546}$$

$$T = 273 \text{ K} = 0^\circ \text{C}$$

28. Ans (2)

$$\text{Here, } C_V = \frac{3}{2}R$$

$$\text{Since } C_P - C_V = R$$

$$\therefore C_P = C_V + R = \frac{3}{2}R + R = \frac{5}{2}R$$

29. Ans (2)

$$\begin{aligned} \gamma_{\text{mix}} &= \frac{n_1 C_{P1} + n_2 C_{P2}}{n_1 C_{V1} + n_2 C_{V2}} \\ &= \frac{1 \times \frac{5}{2}R + 1 \times \frac{7}{2}R}{1 \times \frac{3}{2}R + 1 \times \frac{5}{2}R} = \frac{12}{8} = \frac{3}{2} \\ &= 1.5 \end{aligned}$$

30. Ans (2)

$$U = \frac{f}{2} nRT$$

$$U = \frac{3}{2} \times 2 \times RT = 3RT$$

32. Ans (2)

Even power will not give zero value.

33. Ans (2)

For polytropic process

$$W = \frac{\mu R \Delta T}{1 - x}$$

$$\text{Here } x = -2$$

$$W = \frac{\mu R (T_2 - T_1)}{1 - (-2)}$$

34. Ans (2)

$$v \propto \sqrt{T}$$

$$\Rightarrow \frac{v}{200} = \sqrt{\frac{400}{300}} \Rightarrow v = \frac{400}{\sqrt{3}} \text{ m/s}$$

35. Ans (3)

High temperature and low density.

Note : All real gas behave as an ideal gas at high temperature, low pressure and low density.

SECTION - B

36. Ans (3)

$$\sin i = \mu_2 \sin r_1$$

$$\sin i = (\mu) \sin(A)$$

$$\therefore r^2 = 0$$

$$r_1 + r_2 = A$$

37. Ans (1)

TG: @Chalnaayaaar

Focal length of lens will increase by four times (i.e. 12 cm) while focal length of mirror will not affected by medium.

38. Ans (4)

Parallel in A & D

$$\text{So } \mu_1 = \mu_4$$

39. Ans (3)

$$i > c$$

$$\Rightarrow i > \sin^{-1} \left(\frac{3}{4} \right)$$

$$\sin C = \frac{n_R}{n_D} = \frac{V_D}{V_R} = \frac{1.5 \times 10^8}{2 \times 10^8} = \frac{3}{4}$$

$$C = \sin^{-1} \left(\frac{3}{4} \right)$$

40. Ans (2)

$$\begin{aligned} \frac{n_2}{v} - \frac{n_1}{u} &= \frac{n_2 - n_1}{R} \\ \Rightarrow \frac{1}{V} - \frac{3/2}{-15} &= \frac{1 - 3/2}{-10} \\ \Rightarrow \frac{1}{V} &= \frac{1}{20} - \frac{1}{10} \\ \Rightarrow V &= -20 \text{ cm} \end{aligned}$$

41. Ans (1)

$$s_1 = \frac{1}{2} at^2 = \frac{1}{2} \times 2 \times 100 = 100$$

$$v = at = 20 \text{ m/s}$$

$$s_2 = vt = 20 \times 30 = 600 \text{ m}$$

$$s_3 = \frac{v^2}{2a_1} = \frac{400}{2 \times 4} = 50, |\vec{a}_1| = 4 \text{ (given)}$$

$$\Sigma s = 750 \text{ m}$$

42. Ans (4)

Paramagnetic materials will be feebly attract, diamagnetic material will be feebly repelled and ferromagnetic material will be strongly attracted.

47. Ans (1)

$$P = kV \Rightarrow PV^{-1} = \text{Constant}$$

$$\text{By using } C = C_V + \frac{R}{1 - x} \text{ (for } PV^x = \text{Constant)}$$

$$\text{We have } C = \frac{3}{2}R + \frac{R}{1 - (-1)} = 2R$$

Therefore heat supplied

$$Q = nC\Delta T = 1 (2R) (2T_0 - T_0) = 2RT_0$$

48. **Ans (4)**

$$Q_1 = \Delta U_1 + W_1$$

$$Q_1 = 16 - 20 = -4 \text{ kJ}$$

$$Q_2 = \Delta U_2 + W_2$$

$$W_2 = Q_2 - \Delta U_2 \quad (\because \Delta U_1 = \Delta U_2)$$

$$= 9 - (-4) = 13 \text{ kJ}$$

49. **Ans (4)**

$$PT = \text{constant compare } \frac{T^n}{P^{n-1}} = \text{constant}$$

$$\Rightarrow PT^{\frac{n}{1-n}} = \text{constant}$$

$$\frac{n}{1-n} = 1 \text{ then } n = \frac{1}{2}$$

$$C_{\text{poly}} = C_V + \frac{R}{1-n}$$

$$\Rightarrow 4.5R = \frac{f}{2}R + \frac{R}{1-\frac{1}{2}}$$

$$\Rightarrow f = 5$$

50. **Ans (3)**

$$PV^{2/3} = \text{constant}$$

$$\text{Put } P = \frac{nRT}{V}$$

$$\left(\frac{nRT}{V} \right) V^{2/3} = \text{constant}$$

$$TV^{\frac{2}{3}-1} = \text{constant}$$

$$T \propto V^{\frac{1}{3}}$$

$$\frac{T_1}{T_2} = \left(\frac{V_1}{V_2} \right)^{\frac{1}{3}}$$

$$\frac{T_0}{T_2} = \left(\frac{V_0}{8V_0} \right)^{1/3}$$

$$T_2 = 2T_0$$

SUBJECT : CHEMISTRY

SECTION-A

51. **Ans (1)**

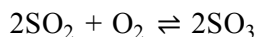
According to question

58. **Ans (4)**

Answer (4)

In Kjeldahl's method nitrogen evolves as NH_3 .

65. **Ans (2)**



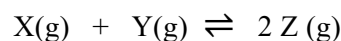
$$Q_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 \times [\text{O}_2]}$$

$$= \frac{(10^{-3})^2}{(10^{-3})^2 \times (10^{-3})} = 10^3$$

$$\because Q_c > K_c$$

$\therefore$ Equilibrium will shift in backward direction.

66. **Ans (3)**



$$t = 0 \quad 0.2 \text{ M} \quad 0.1 \text{ M} \quad 0.1 \text{ M}$$

$$t_{\text{eq}} \quad (0.2 - x) \quad (0.1 - x) \quad (0.1 + 2x)$$

$$\because [\text{Z}] = (0.1 + 2x) = 0.2$$

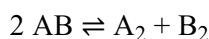
$$\Rightarrow x = 0.05$$

$$\text{Now, } K_c = \frac{[\text{Z}]^2}{[\text{X}]^1 [\text{Y}]^1}$$

67. **Ans (2)**

Additional of inert gas at constant pressure will increase volume so reaction moves in the direction having more gaseous moles.

68. **Ans (2)**



$$K = \frac{x^2}{4(1-x)^2} = \frac{1}{64}$$

$$\frac{x}{2(1-x)} = \frac{1}{8}$$

$$\frac{x}{1-x} = \frac{1}{4}$$

$$4x = 1 - x$$

$$5x = 1$$

$$x = \frac{1}{5} = .2$$

or 20 %

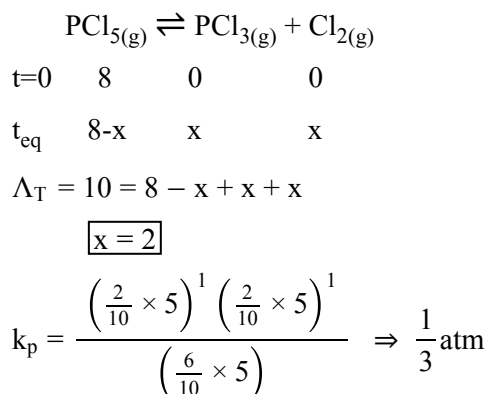
69. **Ans (3)**

$$K_p = 2 \times 6 = 12 \text{ atm}^2$$

70. **Ans (2)**

$$K_c = \frac{[\text{NaCN}]}{[\text{HCN}][\text{NaOH}]} \Rightarrow \frac{[\text{M}]}{[\text{M}][\text{M}]} = \text{M}^{-1}$$

71. Ans (3)



72. Ans (3)

$$\begin{aligned}
 [\text{H}^+] &= \sqrt{k_w} \\
 [\text{H}^+] &= \sqrt{4 \times 10^{-14}} = 2 \times 10^{-7} \\
 \text{pH} &= 7 - \log 2 \Rightarrow 6.7, \text{ Neutral}
 \end{aligned}$$

73. Ans (1)

$$(\text{g eq})_{\text{acid}} > (\text{g eq})_{\text{base}}$$

74. Ans (4)

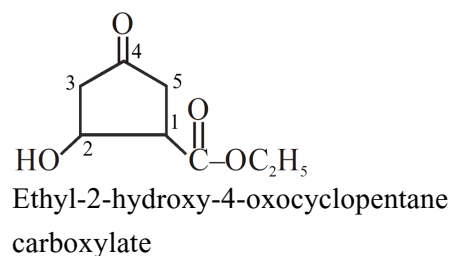
$\ddot{\text{F}} - \ddot{\text{F}}$ bond is weaker

75. Ans (3)

XeF_4 have polar bond and l.p. at C. A.

SECTION-B

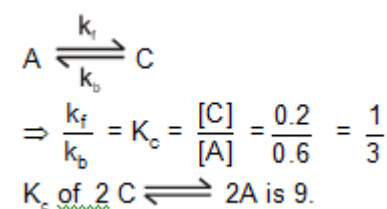
86. Ans (2)



92. Ans (2)

Alc to Le chatelier principle if pressure is applied then equilibrium will shift in that direction where density of substance is more

93. Ans (2)



94. Ans (2)

TG: @Chalnaayaaar

$$\alpha \propto (v)^{-\frac{1}{2}} \rightarrow \Delta_{ng}$$

sum of stoichiometric coefficient of gaseous product.

96. Ans (3)

same I effect $\text{O} > \text{P} (\mu)$
Opp I effect $\text{P} > \text{O} (\mu)$

97. Ans (4)

$\text{SO}_4^{2-}, \text{CO}_3^{2-}, \text{OH}^-$ stability increases downside.

98. Ans (2)

NCERT Pg # 121
Orbitals have almost equal energy.

99. Ans (2)

Atomic orbital is monocentric and molecular orbital.

SUBJECT : BIOLOGY-I

SECTION-A

101. Ans (4)

12th NCERT Pg.208

102. Ans (1)

12th NCERT Pg.209

103. Ans (3)

12th NCERT Pg.212

104. Ans (3)

12th NCERT Pg.171

105. Ans (1)

11th NCERT Pg.146,147

106. Ans (2)

11th NCERT Pg.146 Table 9.3

107. Ans (1)

12th NCERT Pg.209

108. Ans (2)

12th NCERT Pg.213

109. Ans (3)

11th NCERT Pg.150

110. Ans (1)

11th NCERT Pg.148

111. Ans (2)

12th NCERT Pg.214

112. **Ans (1)**
12th NCERT Pg.210, 211
113. **Ans (4)**
NCERT-XII, Pg. # 5
114. **Ans (2)**
NCERT-XII, Pg. # 5
115. **Ans (2)**
NCERT-XII, Pg. # 21
116. **Ans (4)**
NCERT-XII, Pg. # 21
117. **Ans (4)**
NCERT-XII, Pg. # 28
118. **Ans (2)**
NCERT-XII, Pg. # 28
119. **Ans (4)**
NCERT-XII, Pg. # 27
120. **Ans (3)**
NCERT-XII, Pg. # 30
121. **Ans (1)**
NCERT-XII, Pg. # 36, 37
125. **Ans (1)**
NCERT-XII, Pg. # 27
126. **Ans (4)**
NCERT-XII, Pg. # 23
127. **Ans (4)**
NCERT-XII, Pg. # 23
128. **Ans (2)**
NCERT-XII, Pg. # 22
129. **Ans (4)**
NCERT [XII] Pg # 25, Fig. - 2.7 (d)
130. **Ans (4)**
NCERT [XII] Pg # 30
131. **Ans (2)**
NCERT [XII] Pg # 31

SECTION-B

136. **Ans (4)**
12th NCERT Pg.211, 213
137. **Ans (2)**
12th NCERT Pg.177

138. **Ans (4)**
11th NCERT Pg.145
139. **Ans (3)**
12th NCERT Pg.173,174
140. **Ans (4)**
11th NCERT Pg.147,148,149
141. **Ans (4)**
NCERT-XII, Pg. # 5
142. **Ans (3)**
NCERT-XII, Pg. # 27
143. **Ans (1)**
NCERT-XII, Pg. # 29
144. **Ans (2)**
NCERT-XII, Pg. # 30, 31
145. **Ans (4)**
NCERT-XII, Pg. # 21
146. **Ans (3)**
NCERT-XII, Pg. # 22, 23
147. **Ans (3)**
NCERT-XII, Pg. # 25, Fig. 2.7

SUBJECT : BIOLOGY-II
SECTION-A

156. **Ans (4)**
NCERT-XI, Page # 169, (Fig. 10.3), 166 (Fig. 10.2)
160. **Ans (2)**
NCERT-XI, Pg # 131
162. **Ans (2)**
NCERT-XI, Pg#135
164. **Ans (3)**
NCERT XI, Page # 168, 10.41
167. **Ans (4)**
NCERT XI Pg. # 163

SECTION-B

192. **Ans (4)**
NCERT XI, Page # 140, Summary